Setting	Method	Asia		Sachs		Alarm		Insurance		Hailfinder		SIIM-Pn.	
		T	C. Cov.	T	C. Cov.	T	C. Cov.	T	C. Cov.	T	C. Cov.	T	C.Cov.
Cent. (Upper Bound)	OpaqNN	80.6±2.0	100	77.5±2.1	100	73.6±1.5	100	76.6±2.0	100	73.2±0.8	100	75.5±0.8	100
	CBM	79.9±2.0	100	77.2±1.7	100	73.7±1.2	100	76.4±2.0	100	75.8±1.5	100	75.1±0.6	100
	CEM	80.6±2.0	100	77.6±1.7	100	70.7±0.6	100	76.2±2.2	100	72.6±1.9	100	75.1±0.4	100
	CGM	79.3±2.2	100	76.5±1.7	100	70.4±1.2	100	73.2±1.9	100	72.4±1.8	100	69.5±2.2	100
	C^2BM	80.1±2.0	100	77.4±1.8	100	72.0±1.2	100	75.1±2.0	100	74.1±1.8	100	74.4±0.6	100
Loc.	OpaqNN	62.9±8.8	43.3±4.1	55.1±16.1	56±4.0	36.1±18.2	40.9±7.1	19.2 ± 12.3	70.5±7.4	44.7±15.2	63.7±3.5	61.5±4.7	37.5±0.4
	CBM	48.7±6.4	43.3±4.1	45.9±20.9	56±4.0	45.9±11.0	40.9±7.1	31.2±15.7	70.5±7.4	45-4±13.3	63.7±3.5	59.2±11.1	37.5±0.4
	CEM	65.8±15.0	43.3±4.1	57.3±14.4	56±4.0	36.6±15.8	40.9±7.1	39.0±10.8	70.5±7.4	43.3±17.8	63.7±3.5	65.2±2.1	37.5±0.4
	CGM	67.4±8.4	43.3±4.1	50.5±16.0	56±4.0	36.6±17.5	40.9±7.1	32.1±14.1	70.5±7.4	38.1±8.8	63.7±3.5	63.1±0.1	37.5±0.4
	C^2BM	61.9±19.2	43.3±4.1	43.2±24.6	56±4.0	36.1±17.9	40.9±7.1	22.2± 23.3	70.5±7.4	36.7±7.8	63.7±3.5	63.1±0.1	37.5±0.4
External Static FL	FCL	60.1±15.1	N/A	63.2±2.2	N/A	42.3±5.0	N/A	62.2±3.5	N/A	31.5±5.3	N/A	60.2±3.7	N/A
	FedCBM	63.2±11.0	43.3±8.0	57.3±19.4	56.0±7.8	46.9±16.5	68.7±11.9	59.7±17.1	83.2±5.1	49.9±13.3	84.7±5.5	64.9±7.3	65.0±9.2
S-F-CMs	OpaqNN	71.6±16.2	43.3±8	59.8±19.3	56±7.8	49.3±20.5	68.7±11.9	64.9±14.7	83.2±5,1	50.7±19.9	84.7±5.5	65.1±2.5	65±9.2
	CBM	59.9±11.0	43.3±8	50.9±26.1	56±7.8	41.3±21.0	68.7±11.9	58.5±23.3	83.2±5.1	57.3±17.9	84.7±5.5	62.9±12.8	65±9.2
	CEM	68.1±14.7	43.3±8	55.0±26.2	56±7.8	44.1±23.1	68.7±11.9	68.8±6.4	83.2±5.1	51.8±21.2	84.7±5.5	64.7±2.4	65±9.2
	CGM	67.1±14.4	43.3±8	44.0±28.8	56±7.8	52.6±18.5	48.7±20.6	57.9±23.0	76.8±11.6	42.9±14.8	84.7±5.5	63.2±0.2	65±9.2
	C^2BM	59.7±20.2	43.3±8	59.8±26.5	56±7.8	43.5±23.1	48.7±20.6	63.0±21.5	76.8±11.6	54.5±21.8	84.7±5.5	62.7±12.7	65±9.2
F-CMs	OpaqNN	**80.0±1.8**	**100**	**75.1±2.1**	**100**	**72.9± 0.5**	**100**	**71.8±2.3**	**100**	**66.0±3.4**	**100**	**67.1±2.4**	**100**
	CBM	**79.8±1.4**	**100**	**75.3± 2.0**	**100**	**71.5±3.8**	**100**	**69.5± 1.1**	**100**	**72.4±3.6**	**100**	**71.0**±0.2	**100**
	CEM	**80.5±1.7**	**100**	**76.0±2.6**	**100**	**73.4±0.4**	**100**	**71.0± 0.8**	**100**	**70.2±3.8**	**100**	**67.3±2.7**	**100**
	CGM	**80.0±2.0**	**100**	**70.2±3.2**	**100**	**69.1±5.9**	**100**	**70.8± 2.3**	**100**	**55.5±2.9**	**99.5±0.5**	**66.1±1.9**	**100**
	C^2BM	**80.4± 2.2**	**100**	**76.2±2.1**	**100**	**73.1±0.8**	**100**	**74.3± 2.3**	**100**	**72.8±2.1**	**99.5±0.5**	**68.8±2.5**	**100**

Table 1: **Extended Table 1 of the manuscript, including additional baselines.** *Task accuracy* (%) and *concept coverage* (%) with respect to the ground-truth concepts causally relevant for the task. Methods that improve the average performance are shown in bold.

Data	Model	d_inst.	d_par.	agg.	n_rounds	tot.
Asia	F-CMs (CEM)	0.227 ± 0.001	0.002 ± 0.000	–	42.583 ± 2.256	229.726 ± 13.609
SIIM	F-CMs (CEM)	0.691 ± 0.093	0.003 ± 0.000	–	35.350 ± 5.188	506.230 ± 75.622
Asia	F-CMs (C2BM)	0.244 ± 0.001	0.003 ± 0.000	0.063 ± 0.081	61.500 ± 3.679	370.800 ± 24.924
SIIM	F-CMs (C2BM)	0.672 ± 0.102	0.004 ± 0.000	0.037 ± 0.003	56.550 ± 7.278	922.543 ± 120.707

Table 2: **Calculation of average time (in seconds) for the different steps of our methodology**, reported as mean ±2 standard deviation across three runs. Specifically, *d_inst*, the time spent instantiating the architecture after client drift; *d_par*, the time spent initializing the post-drift architecture with the old learned parameters; *agg*, the time spent aggregating client graphs; *n_rounds*, the number of training rounds; and *tot.*, the total training time. These results are consistent with our complexity analysis, showing that the additional overhead introduced by F-CMs is modest and primarily due to lightweight, event-triggered operations, remaining small compared to the dominant cost of model training.

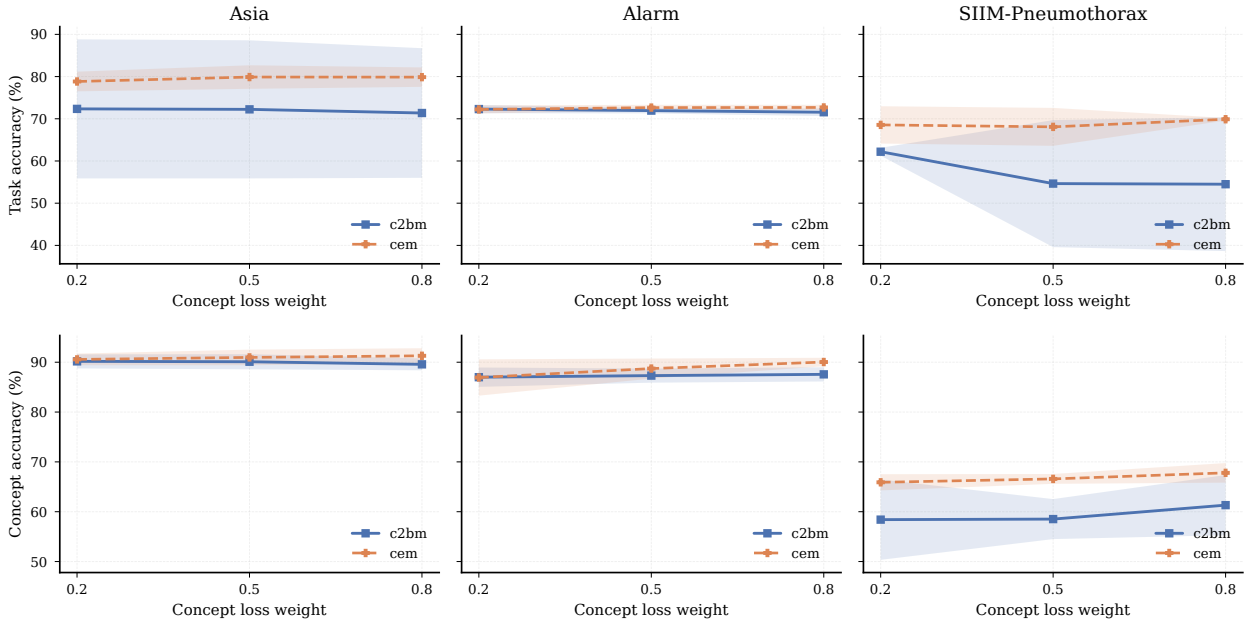

Figure 1: **Sensitivity analysis on γ.** *Task accuracy* (%) and *concept accuracy* (%) as a function of the concept loss weight (γ) for the C2BM and CEM instantiations across the Asia, Alarm, and SIIM-Pneumothorax datasets, calculated over three runs. The results show that downstream performance remains robust to changes in the concept-task trade-off, even on datasets with a larger number of concepts. A slight upward trend is observed in concept accuracy on SIIM-Pneumothorax as γ increases, which is expected since the model places greater weight on the concept prediction loss, accompanied by a corresponding slight downtrend in task accuracy.

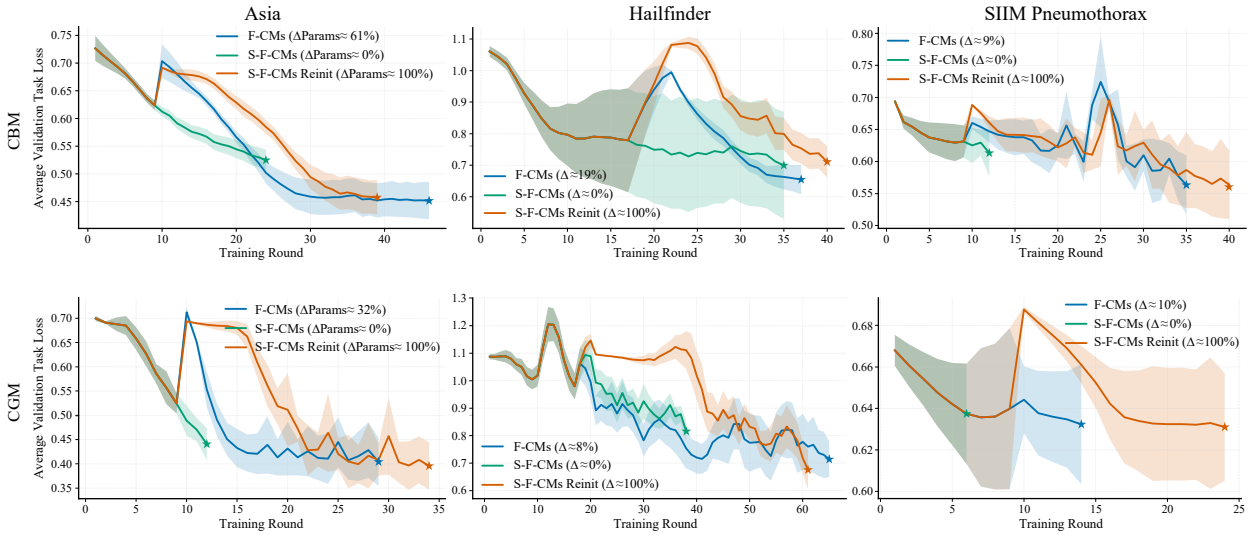

Figure 2: **Additional convergence under temporal non-stationarity of CBM and CGM.** Average validation task loss across rounds for CBM and CGM instantiations, where new clients introduce unseen concepts. F-CMs (ours) adapt the architecture modularly and generally converge faster and/or to lower loss than S-F-CMs (no adaptation) and S-F-CMs Reinit (full retraining). The legend reports the fraction of modified parameters.

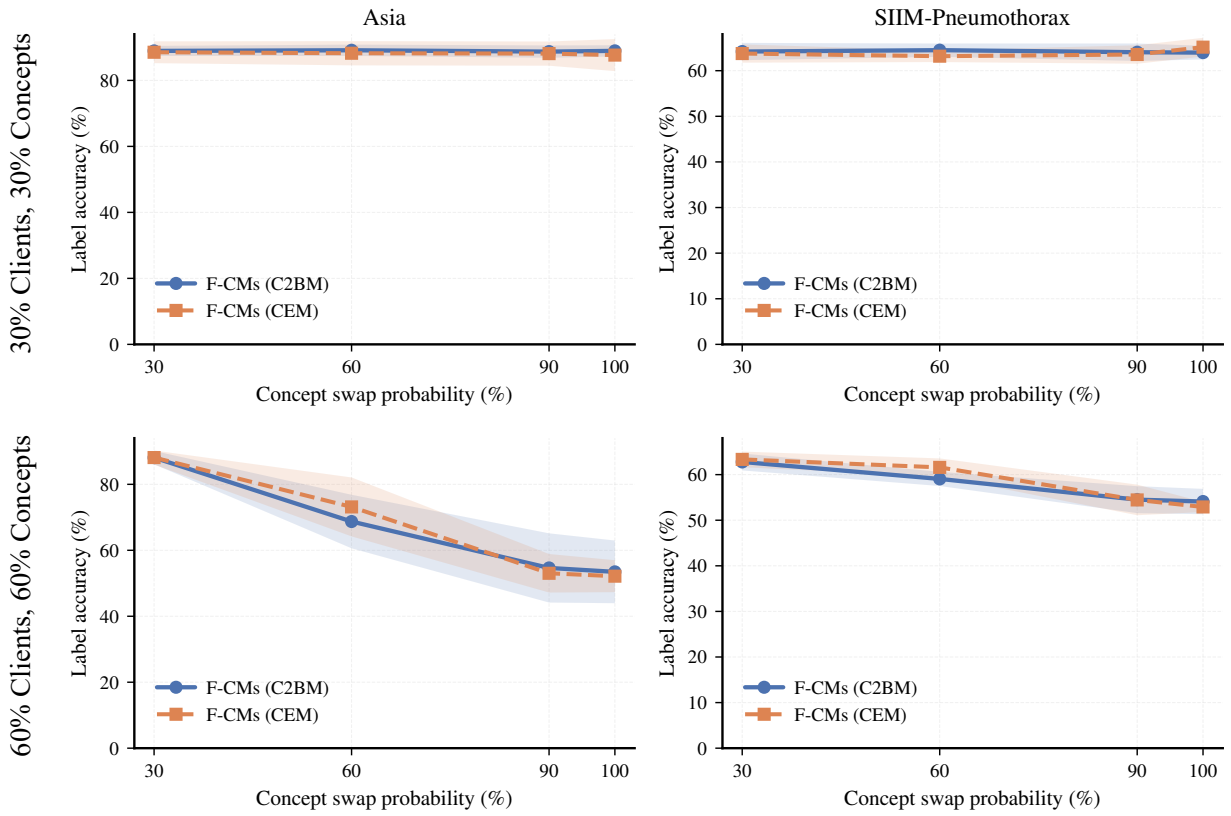

Figure 3: **Ablation study on inter-client semantic disagreement.** *Label accuracy (%)* of two representative F-CM instantiations, bipartite (F-CMs (CEM)) and graph-based (F-CMs (C2BM)), under varying fractions of concept label swaps within clients during training. Each row shows a configuration with either 30% of concepts and 30% of clients, or 60% of concepts and 60% of clients selected for swapping. For each concept, the corresponding fraction of clients is selected among those with supervision on it, and within them a fraction of the concept labels (x-axis: 30%, 60%, 90%, 100%) is swapped, simulating conflicting definitions across clients. Label accuracy is evaluated on a test set where concepts are not swapped; values are reported over three runs. Each column corresponds to a different dataset.

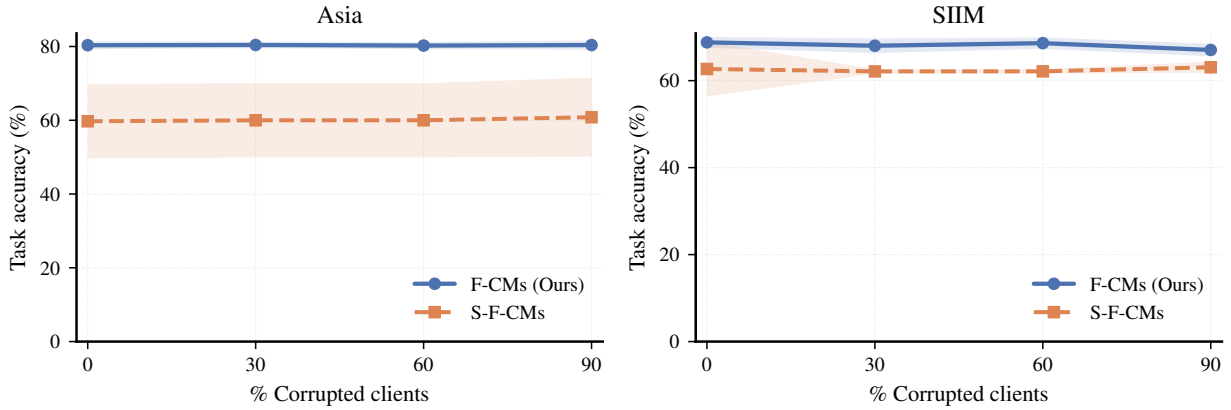

Figure 4: **Ablation study on downstream task robustness to graph misspecification.** Task accuracy of C2BM instantiations under the same perturbation setting as Figure 9, with graph alteration $p = 0.6$, shown as a function of the percentage of corrupted clients. On both Asia and SIIM-Pneumothorax, accuracy remains largely stable despite structural degradation, suggesting limited downstream sensitivity to moderate errors in the aggregated graph, while F-CMs consistently operates at a higher accuracy level than S-F-CMs.

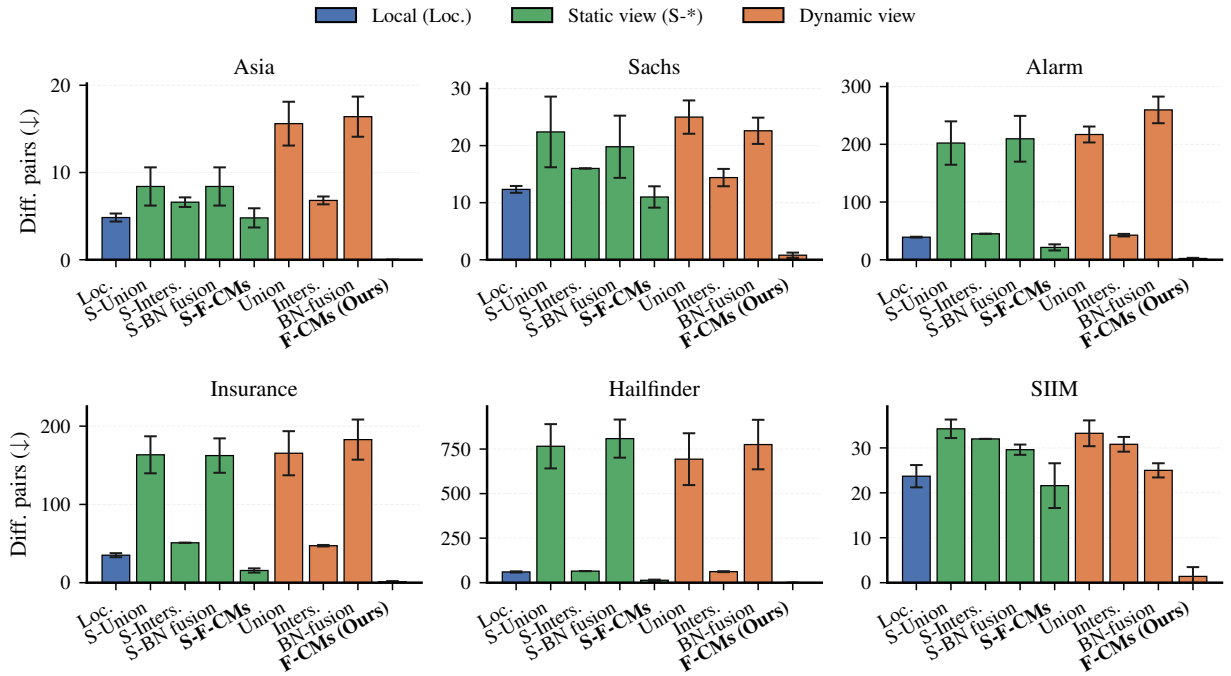

Figure 5: **Ablation of graph aggregation strategies under moderate perturbations.** DiffPairs (lower is better) for different aggregation rules under localized, static federated, and dynamic federated visibility, with graph alteration $p = 0.3$ and 30% corrupted clients.

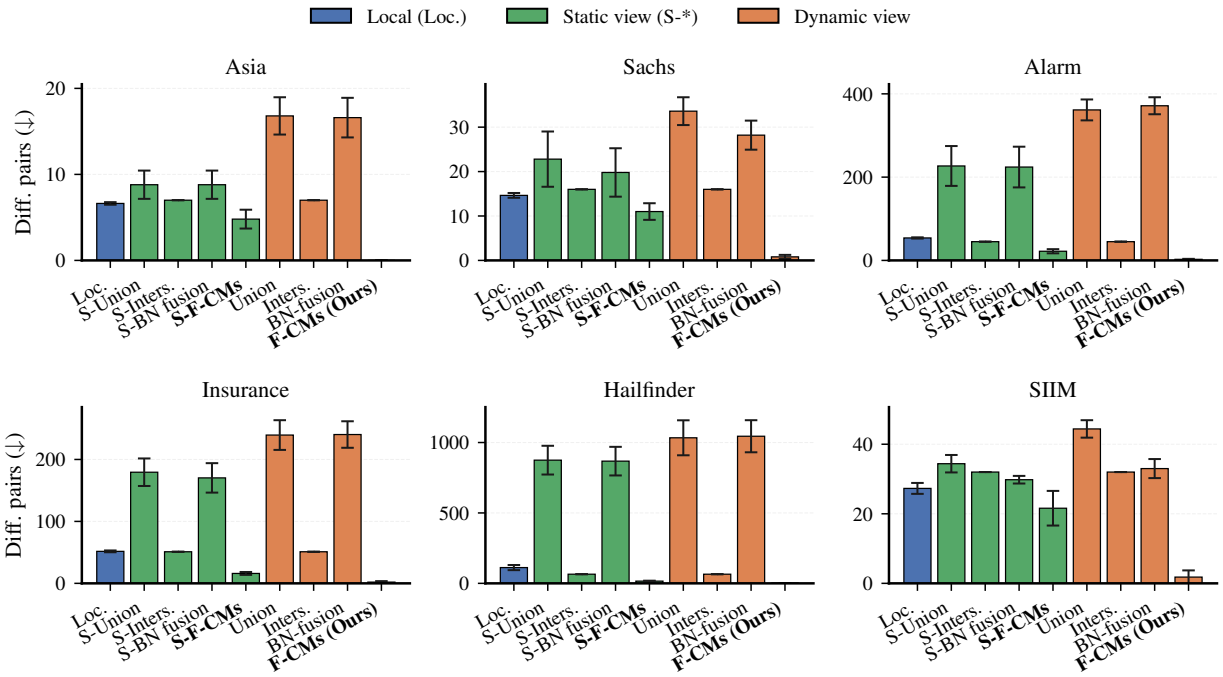

Figure 6: **Ablation of graph aggregation strategies under stronger perturbations.** DiffPairs (lower is better) for different aggregation rules under localized, static federated, and dynamic federated visibility, with graph alteration $p = 0.6$ and 60% corrupted clients.